



New liver cell

Using single-cell RNA sequencing, creation of liver tissue is facilitated which could end the need for transplants. <https://digit.in/sep19-13>



Chocolate prevents depression?

A UCL-led study found that eating more dark chocolate may positively affect mood and relieve depression. <https://digit.in/sep19-14>



SMARTPHONE MIND CONTROL

No, not the kind of mind control you're thinking about. This innovative device can control brain cells using a simple smartphone app!

Dhriti Datta | dhriti@digit.in

W

e have all imagined owning devices that can control the human mind. While such expectations

may border upon science-fiction, there have been some innovations that have brought us closer to this ethereal possibility. Companies such as Microsoft and Facebook are tirelessly working towards actual 'mind-control' brain interfaces, with Microsoft even filing a patent for their technology in the same space. Samsung has also

revealed that it is working towards a smartphone that can exclusively be controlled by a user's brainwaves. We are at least a decade away from such advanced technological models becoming mainstream except for this one group of researchers whose innovation has already begun testing. Their aim? To manipulate brain cells wirelessly using a smartphone.

The group is formed of engineers hailing from the Korea Advanced Institute of Science and Technology (KAIST) and a team of neuroscientists from the University of Washington, and they recently published their research in the Nature Biomedical Engineering

journal. The research discusses their novel innovation - a device that can control the neural circuits in a human brain using a tiny brain implant which is controlled by a smartphone. This is a commendable innovation in the field of neuropsychiatry, which is the branch of medicine that treats mental health problems such as migraines, anxiety, depression, ADHD, schizophrenia, bipolar disorder, Parkinson's diseases, and many more. Introducing a new drug, right from discovery to market, is often a costly and arduous process in this field. The device created by KAIST and neuroscientists from the University of Washington paves a new path for drug delivery and optogenetics that works over extended periods of time.

This device can also be utilised to extract liquids from the brain which could aid in diagnostic sciences, as per Raza Qazi, lead author of this study